MIL-E-1/1404C
12 September 1972
SUPERSEDING
MIL-E-1/1404B(EL)
29 April 1964

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 2DF4

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pentode, miniature, power amplifier (100 MHz, Class C)

Outline --- 6-2(EIA)

Base --- E9-1

Envelope --- T6-1/2

Cathode --- Coated filament

Mounting --- See note 1

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	g2	int con	a	f- (+prl)	f+	nc	int con	g1	g3, fct (-prl)

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ik	Ic1	Pa	Pg2	Alt
Unit:	Vdc	Vdc	Vdc	Vdc	mAdc	mAdc	W	W	ft
Maximum:	1.44 or 2.88 (prl) (ser)	250	-75	125	50	4	4.5	1.5	Note 2
Minimum:	1.06 or 2.12 (prl) (ser)	---	---	---	---	---	---	---	---
TEST CONDITIONS:	2.5 (ser)	120	-3.6	120	---	---	---	---	---

GENERAL:

© First article testing - Required (see note 11)

© denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1266	Total grid current	4		0.65	II	Ic1	0	-1.0	µAdc
1256	Electrode current (1) (anode)	4		0.65	II	Ib	28	52	mAdc
---	Operation current (1)	5		0.65	II	Ip	60	72	mAac
---	Operation current (2)	5	Ef = 2.125 Vdc	0.65	II	Ip	55	---	mAac
1201	Short and discontinuity detection	-		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1256	Electrode current (screen)	-		---	---	Ic2	0	7.0	mAdc
1256	Electrode current (2) (anode)	-	Ec1 = -20 Vdc	---	---	Ib	---	2.0	mAdc
1316	Amplification factor (triode)	-	Screen grid tied to anode	---	---	Mu	8	14	---
1266	Primary grid emission (screen)	6	If = 2.5 Vac; Eb = 0; Ec2 = 150 Vac; t = 300 (max)	---	---	Is	---	200	µAdc
Ⓒ ---	Heating time	7	Ef = 2.5 Vac; Ebb = 150 Vdc; Ecc2 = 120 Vdc; Ecc1 = 0; Rg1 = 20,000 ohms; Eg1 = 13 Vac; Rf = 1,000 ohms; Rg2 = 470 ohms; Ib = 17 mAdc; C1 = 16 µF (see figure 2)	---	---	t	---	0.5	sec
1331	Direct-interelectrode capacitance	-	No shield	---	---	Cgp Cin Cout	--- 5.8 3.9	0.25 8.8 6.9	pF pF pF
1236	Power oscillation (1)	-	F = 70 MHz; Ef = 2.5 Vdc; Rf = 2,000 ohms; Ecc1 = 0; Eg1 = 13 Vac; Rg2 = 1,800 ohms; Rg1 = 3,300 ohms	---	---	Po Ic1	2.0 0.2	--- 3.0	W mAdc
1236	Power oscillation (2)	-	F = 70 MHz; Ef = 2.125 Vdc; Rf = 2,000 ohms; Ecc1 = 0; Rg1 = 3,300 ohms; Rg2 = 1,800 ohms; Eg1 = 13 Vac	---	---	Po Ic1	1.5 0.2	--- 3.0	W mAdc

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued								
1236	Power oscillation (3)	-	F = 100 MHz; Ef = 1.25 Vdc; Rf = 2,500 ohms; Ecc1 = 0; Ebb = 150 Vdc; Ecc2 = 90 Vdc; Rg1 = 15,000 ohms; Eg1 = 37 Vac	---	---	Po	1.5	---	W
1031	Low-frequency (25 Hz) vibration	-	Rp = 2,000 ohms; Rg1 = 0.1 Meg	---	---	Ep	---	500	mVac
---	Resonance	8	Rp = 2,000 ohms; Rg1 = 0.1 Meg	---	---	Ep	---	3,800	mVac
1041	Shock	-	450 G; Ef = 2.5 Vac (no other voltages applied)	---	---	---	---	---	---
---	Post-shock test end points:								
1301	Filament current (1)	-		---	---	If	300	---	mAdc
1256	Electrode current (1) (anode)	-		---	---	Ib	26	---	mAdc
1266	Total grid current	-		---	---	Ic1	---	-1.5	µAdc
1306	Transconductance	-		---	---	Sm	5,300	---	µmhos
---	Vibration fatigue	3, 9	No voltages applied	---	---	---	---	---	---
---	Post-vibration fatigue test end points:								
1301	Filament current (1)	-		---	---	If	300	---	mAdc
1256	Electrode current (1) (anode)	-		---	---	Ib	26	---	mAdc
1266	Total grid current	-		---	---	Ic1	---	-1.5	µAdc
1306	Transconductance	-		---	---	Sm	5,300	---	µmhos
(C) 1301	Filament current (1)	-	Ef = 2.5 Vdc (series connected)	1.0	S3	If	320	370	mAdc
1301	Filament current (2)	-	Ef = 1.25 Vdc (parallel connected)	4.0	S3	If	640	740	mAdc
1306	Transconductance	-		4.0	S3	Sm	5,800	9,200	µmhos
1121	Base strain	-		---	---	---	---	---	---
2126	Glass strain	-		2.5	I	---	---	---	---
(C) 1211	Insulation of electrodes	-		4.0	S3	---	---	---	---
(C) 1105	Permanence of marking	-		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC		SYMBOL	LIMITS		UNIT
					1ST SAMPLE	COMBINED SAMPLES		MIN	MAX	
	Quality conformance inspection, part 3									
1506	Heater-cycling life	Ef = 3.0 Vac (series connected)	---	---	---	---	---	5,000	---	Cycles
---	Heater-cycling life-test end point:									
1301	Filament current (1)		---	---	---	---	If	320	370	mAdc
1501	Intermittent life	Ef = 2.5 Vac; F = 60 Hz; Ebb = 150 Vdc; Ecc2 = 125 Vdc; Ecc1 = -10 Vdc; Rf = 1,000 ohms; Rg2 = 470 ohms; Eg1 = 13 Vac; Rg1 = 4,700 ohms; C1 = 16 μF (see note 10 and figure 2)	---	---	---	---	---	---	---	---
---	Intermittent life-test end points (500 hours):									
---	Inoperatives		---	---	1	3	---	---	---	---
---	Operation current (1)		---	---	1	3	Ip	47	---	mAac
1338	Power oscillation (2)		---	---	1	3	Po	1.4	---	W
---	Total defectives		---	---	2	4	---	---	---	---
(C)	Periodic-check test									
1202	Filament burnout		---	---	---	---	---	---	---	---

NOTES:

1. Vertical mounting is preferred. For horizontal mounting, pins No. 4 and No. 5 shall be in the same horizontal plane.
2. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
3. This is essentially a variable-frequency vibration test. The tubes under test shall be mounted rigidly on a table vibrating, at simple harmonic motion, over a frequency range of 10 to 2,000 Hz. The amplitude shall be kept constant at 0.04 inch (1.00 mm), from 10 to 55 Hz; and, from 55 to 2,000 Hz, the peak acceleration shall be kept constant at 10 G. Each tube shall be subjected to vibration cycles in orientations X, Y, and Z. The time for one complete cycle: 10 to 2,000 to 10 Hz, shall be 6 minutes ±1 minute. The time of vibration for each tube in each of the three orientations shall be 30 minutes ±5 minutes.
4. This test to be performed at the conclusion of the holding period.
5. Test at power line frequency (see figure 1). Tubes shall not be preheated immediately before this test.
6. (C) With the voltages applied as specified, adjust Ec1 to provide Ic2 = 9.0 mAdc on the positive half-cycles.
7. The time, from the application of the filament voltage until the anode current attains the minimum value specified, shall be within the time limit specified. (A relay is switched into the anode circuit, which stops the timer when the anode current reaches predetermined value. The timer starts with application of filament voltage.)

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NOTES: -Continued

8. This is essentially a sweep-frequency vibration test. The tube under test shall be mounted on a vibration table vibrating at simple harmonic motion. Low-frequency (25 Hz) vibration test conditions shall be applied, and Ep monitored while the frequency of vibration is continuously swept from 50 to 5,000 Hz and back, with the peak acceleration controlled constant at 2 G. Total time for each sweep cycle shall be not less than 1 minute.

9. This test shall be conducted on the initial lot submitted under a contract or order, and shall be conducted again after 12 months of continued production under a contract or order.

10. Group A, life-test sampling plan, except that the following sampling and acceptance criteria, shall apply:

Double sampling plan

	Sample size	Cumulative Sample size	Acceptance No.	Rejection No.
1st sample.....	10	10	1 inoperative or 2 total defectives.	4 inoperatives or 5 total defectives.
2nd sample.....	20	30	3 inoperatives or 4 total defectives.	4 inoperatives or 5 total defectives.

Criteria for prerelease

Eligibility at	Allowable number of defectives at 1/				
	Tube sample	10% of specified life	30% of specified life	60% of specified life	100% of specified life
10% of specified Life.....	Current 10	0	.....	.....	.....
	Last 60	1	3	5	8
30% of specified life	Current 10	.....	0	.....	.....
	Last 60	.....	5	6	8
60% of specified life	Current 10	.....	.....	1	.....
	Last 60	.....	.....	6	8

1/ The requirements of this tabulation are applicable to first sample only.

More rapid establishments for eligibility for prerelease:

Larger life-test samples may be selected from initial life-test lots in accordance with the following:

(a) The number of tubes from any lot shall not exceed 20.

(b) In those instances where more than 10 tubes are selected, the tubes shall be subsequently numbered and the first 10 tubes shall be considered the first sample for evaluation of the lot they represent.

11. First article inspection shall consist of performing all tests listed on this TSS. The sample size and allowable defects shall be in accordance with MIL-E-1, Qualification, Samples and acceptance criteria. Three copies of the test report, prepared in accordance with MIL-STD-813, shall be forwarded to the purchasing activity.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES
Other - NS

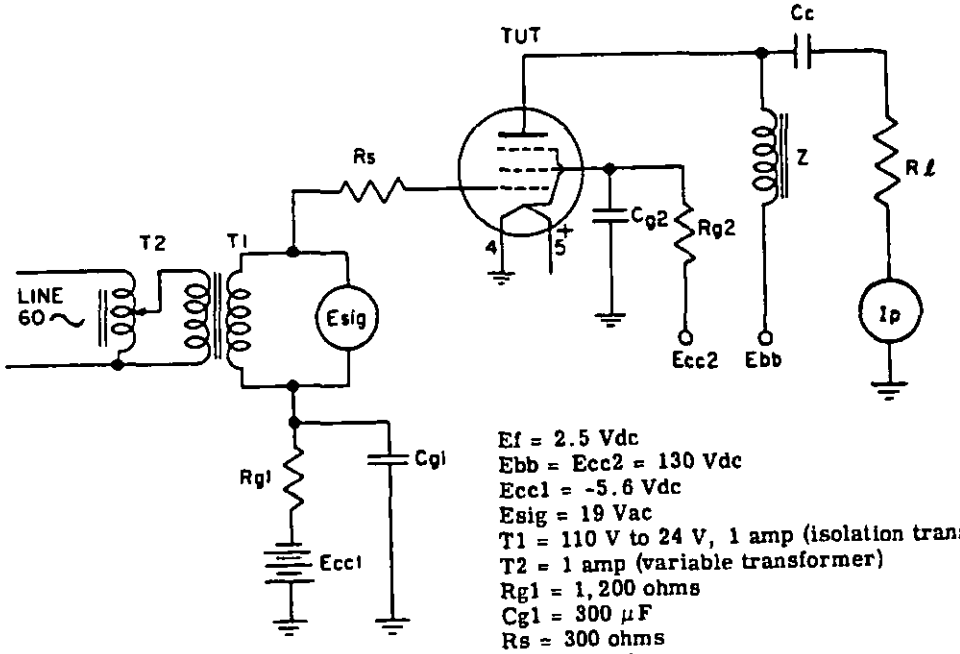
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2619)

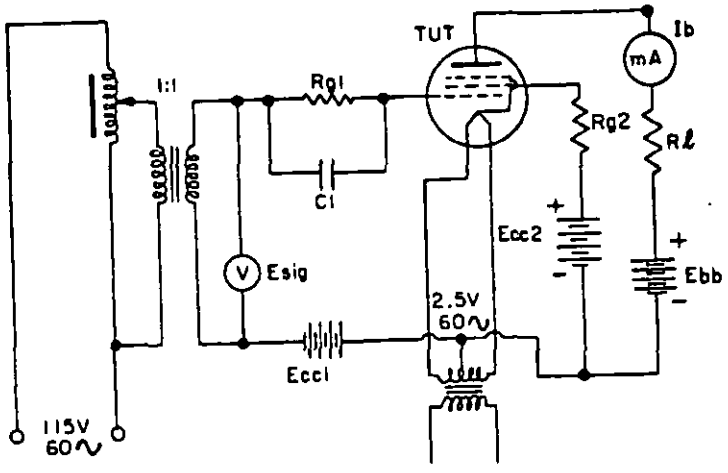
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Ef = 2.5 Vdc
Ebb = Ecc2 = 130 Vdc
Ecc1 = -5.6 Vdc
Esig = 19 Vac
T1 = 110 V to 24 V, 1 amp (isolation transformer)
T2 = 1 amp (variable transformer)
Rg1 = 1,200 ohms
Cg1 = 300 μF
Rs = 300 ohms
Rg2 = 2,200 ohms
Cg2 = 80 μF
Rℓ = 500 ohms (meter plus resistor)
Cc = 20 μF (oil filled)
Z = 60 H (200 ohms)

NOTE: Ip meter shall read RMS current and reactance shall be less than 100 ohms.

FIGURE 1. Operation current test circuit.



NOTE: The resistance Rℓ shall include the resistance of the anode current indicator.

FIGURE 2. Heating time and intermittent life-test circuit.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/140(C ELECTRON TUBE, RECEIVING TYPE 2DF4

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 68 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED. ESC-FM 1068-68

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